Lab 2: Hidden Markov Models

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Question 1: Build the HMM

```
states <- 1:10
symbols <- 1:10

transProbs <- matrix(0,10,10)
emissionProbs <- matrix(0,10,10)

for(i in 1:10) {
  transProbs[i,i] <- 0.5
  transProbs[i, (i%%10) + 1] <- 0.5
  for(k in -2:2){
    emissionProbs[i, ((i+k-1) %% 10) + 1] <- 0.2
  }
}
```

Question 2: Simulate 100 time steps

```
hmm <- initHMM(States = states, Symbols = symbols, transProbs = transProbs,
               emissionProbs = emissionProbs, startProbs = rep(0.1, 10))
simulation <- simHMM(hmm, 100)
```

Question 3: Filtered, smoothed, and most probable path

```
observations <- simulation$observation

filtered <- rep(0, 100)
smoothed <- rep(0, 100)

logAlpha <- forward(hmm, observations)
logBeta <- backward(hmm, observations)

# filtered
filtered <- apply(logAlpha, 2, function(col){
  col <- exp(col - max(col))
  col/sum(col)
```

```
})

#smoothed
smoothed <- apply(logAlpha + logBeta, 2, function(col){
  col <- exp(col - max(col))
  col/sum(col)
})

#probable path
probablePath <- viterbi(hmm, observations)

#prints
print(filtered)
```

```
##       index
## states   1   2     3         4         5         6         7         8
##     1  0.2 0.0 0.000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     2  0.2 0.4 0.000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     3  0.2 0.4 0.500 0.3636364 0.2666667 0.0000000 0.0000000 0.0000000
##     4  0.0 0.2 0.375 0.6363636 0.7333333 0.5769231 0.2884615 0.2678571
##     5  0.0 0.0 0.125 0.0000000 0.0000000 0.4230769 0.5000000 0.7321429
##     6  0.0 0.0 0.000 0.0000000 0.0000000 0.0000000 0.2115385 0.0000000
##     7  0.0 0.0 0.000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     8  0.0 0.0 0.000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     9  0.2 0.0 0.000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     10 0.2 0.0 0.000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##       index
## states         9        10        11         12        13        14         15
##     1  0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.00000000
##     2  0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.00000000
##     3  0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.00000000
##     4  0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.00000000
##     5  0.5773196 0.2886598 0.1613833 0.08069164 0.1744548 0.1485411 0.07427056
##     6  0.4226804 0.5000000 0.4409222 0.30115274 0.8255452 0.8514589 0.50000000
##     7  0.0000000 0.2113402 0.3976945 0.41930836 0.0000000 0.0000000 0.42572944
##     8  0.0000000 0.0000000 0.0000000 0.19884726 0.0000000 0.0000000 0.00000000
##     9  0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.00000000
##     10 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.00000000
##       index
## states         16         17        18 19  20        21        22         23
##     1  0.00000000 0.00000000 0.0000000  0 0.0 0.0000000 0.3333333 0.41666667
##     2  0.00000000 0.00000000 0.0000000  0 0.0 0.0000000 0.0000000 0.16666667
##     3  0.00000000 0.00000000 0.0000000  0 0.0 0.0000000 0.0000000 0.00000000
##     4  0.00000000 0.00000000 0.0000000  0 0.0 0.0000000 0.0000000 0.00000000
##     5  0.03713528 0.01856764 0.0000000  0 0.0 0.0000000 0.0000000 0.00000000
##     6  0.28713528 0.16213528 0.0000000  0 0.0 0.0000000 0.0000000 0.00000000
##     7  0.46286472 0.37500000 0.0000000  0 0.0 0.0000000 0.0000000 0.00000000
##     8  0.21286472 0.33786472 0.0000000  0 0.0 0.0000000 0.0000000 0.00000000
##     9  0.00000000 0.10643236 0.8067429  1 0.5 0.3333333 0.1666667 0.08333333
##     10 0.00000000 0.00000000 0.1932571  0 0.5 0.6666667 0.5000000 0.33333333
##       index
## states         24         25        26        27        28        29        30
##     1  0.37500000 0.30434783 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
```

```
##     2  0.29166667 0.34782609 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     3  0.08333333 0.19565217 0.7352941 0.3676471 0.0000000 0.0000000 0.0000000
##     4  0.00000000 0.00000000 0.2647059 0.5000000 0.0000000 0.0000000 0.0000000
##     5  0.00000000 0.00000000 0.0000000 0.1323529 0.8269231 0.4526316 0.3115942
##     6  0.00000000 0.00000000 0.0000000 0.0000000 0.1730769 0.5473684 0.6884058
##     7  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     8  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     9  0.04166667 0.02173913 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     10 0.20833333 0.13043478 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##       index
## states        31        32        33        34        35         36        37
##     1  0.0000000 0.0000000 0.0000000 0.0000000 0.1549755 0.50000000 0.0000000
##     2  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.07748777 0.8816936
##     3  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.1183064
##     4  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     5  0.1557971 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     6  0.5000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     7  0.3442029 0.7103659 0.3551829 0.0000000 0.0000000 0.00000000 0.0000000
##     8  0.0000000 0.2896341 0.5000000 0.0000000 0.0000000 0.00000000 0.0000000
##     9  0.0000000 0.0000000 0.1448171 0.8166023 0.0000000 0.00000000 0.0000000
##     10 0.0000000 0.0000000 0.0000000 0.1833977 0.8450245 0.42251223 0.0000000
##       index
## states         38         39 40  41   42    43         44         45         46
##     1  0.00000000 0.0000000  0 0.0 0.00 0.000 0.00000000 0.00000000 0.00000000
##     2  0.44084682 0.3190626  0 0.0 0.00 0.000 0.00000000 0.00000000 0.00000000
##     3  0.50000000 0.6809374  0 0.0 0.00 0.000 0.00000000 0.00000000 0.00000000
##     4  0.05915318 0.0000000  1 0.5 0.25 0.125 0.00000000 0.00000000 0.00000000
##     5  0.00000000 0.0000000  0 0.5 0.50 0.375 0.26666667 0.00000000 0.00000000
##     6  0.00000000 0.0000000  0 0.0 0.25 0.375 0.40000000 0.38461538 0.19230769
##     7  0.00000000 0.0000000  0 0.0 0.00 0.125 0.26666667 0.38461538 0.38461538
##     8  0.00000000 0.0000000  0 0.0 0.00 0.000 0.06666667 0.19230769 0.28846154
##     9  0.00000000 0.0000000  0 0.0 0.00 0.000 0.00000000 0.03846154 0.11538462
##     10 0.00000000 0.0000000  0 0.0 0.00 0.000 0.00000000 0.00000000 0.01923077
##       index
## states        47        48         49        50         51 52 53  54        55
##     1  0.0000000 0.0000000 0.06481481 0.8235294 0.41176471  1  0 0.0 0.0000000
##     2  0.0000000 0.0000000 0.00000000 0.1764706 0.50000000  0  1 0.5 0.3333333
##     3  0.0000000 0.0000000 0.00000000 0.0000000 0.08823529  0  0 0.5 0.6666667
##     4  0.0000000 0.0000000 0.00000000 0.0000000 0.00000000  0  0 0.0 0.0000000
##     5  0.0000000 0.0000000 0.00000000 0.0000000 0.00000000  0  0 0.0 0.0000000
##     6  0.1041667 0.0000000 0.00000000 0.0000000 0.00000000  0  0 0.0 0.0000000
##     7  0.3125000 0.2197802 0.00000000 0.0000000 0.00000000  0  0 0.0 0.0000000
##     8  0.3645833 0.3571429 0.32407407 0.0000000 0.00000000  0  0 0.0 0.0000000
##     9  0.2187500 0.3076923 0.37345679 0.0000000 0.00000000  0  0 0.0 0.0000000
##     10 0.0000000 0.1153846 0.23765432 0.0000000 0.00000000  0  0 0.0 0.0000000
##       index
## states        56         57        58         59         60         61
##     1  0.0000000 0.00000000 0.0000000 0.00000000 0.00000000 0.00000000
##     2  0.1666667 0.08333333 0.0000000 0.00000000 0.00000000 0.00000000
##     3  0.5000000 0.33333333 0.0000000 0.00000000 0.00000000 0.00000000
##     4  0.3333333 0.41666667 0.5000000 0.25000000 0.12500000 0.00000000
##     5  0.0000000 0.16666667 0.3888889 0.44444444 0.34722222 0.25185185
##     6  0.0000000 0.00000000 0.1111111 0.25000000 0.34722222 0.37037037
##     7  0.0000000 0.00000000 0.0000000 0.05555556 0.15277778 0.26666667
```

```
##     8  0.0000000 0.00000000 0.0000000 0.00000000 0.02777778 0.09629630
##     9  0.0000000 0.00000000 0.0000000 0.00000000 0.00000000 0.01481481
##     10 0.0000000 0.00000000 0.0000000 0.00000000 0.00000000 0.00000000
##       index
## states         62         63        64 65 66  67        68        69  70
##     1  0.00000000 0.00000000 0.0000000  0  0 0.0 0.0000000 0.0000000 0.0
##     2  0.00000000 0.00000000 0.0000000  0  0 0.0 0.0000000 0.0000000 0.0
##     3  0.00000000 0.00000000 0.0000000  0  0 0.0 0.0000000 0.0000000 0.0
##     4  0.00000000 0.00000000 0.0000000  0  0 0.0 0.0000000 0.0000000 0.0
##     5  0.12686567 0.07439825 0.0000000  0  0 0.0 0.0000000 0.0000000 0.0
##     6  0.31343284 0.25820569 0.0000000  0  0 0.0 0.0000000 0.0000000 0.0
##     7  0.32089552 0.37199125 0.0000000  0  0 0.0 0.0000000 0.0000000 0.0
##     8  0.18283582 0.29540481 0.6931818  1  1 0.5 0.3333333 0.1666667 0.1
##     9  0.05597015 0.00000000 0.3068182  0  0 0.5 0.6666667 0.5000000 0.4
##     10 0.00000000 0.00000000 0.0000000  0  0 0.0 0.0000000 0.3333333 0.5
##       index
## states         71         72        73        74        75         76        77
##     1  0.00000000 0.30000000 0.4423077 0.6825397 0.5000000 0.32936508 0.0000000
##     2  0.00000000 0.00000000 0.1730769 0.0000000 0.3412698 0.42063492 0.4960630
##     3  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.17063492 0.3910761
##     4  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.00000000 0.1128609
##     5  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     6  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     7  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     8  0.06666667 0.03333333 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     9  0.33333333 0.20000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000
##     10 0.60000000 0.46666667 0.3846154 0.3174603 0.1587302 0.07936508 0.0000000
##       index
## states        78        79        80        81         82        83        84
##     1  0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000
##     2  0.2628651 0.1516854 0.0758427 0.1727605 0.08638026 0.0000000 0.0000000
##     3  0.4700974 0.4229535 0.2873194 0.8272395 0.50000000 0.0000000 0.0000000
##     4  0.2670376 0.4253612 0.4241573 0.0000000 0.41361974 0.6883609 0.3441804
##     5  0.0000000 0.0000000 0.2126806 0.0000000 0.00000000 0.3116391 0.5000000
##     6  0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.1558196
##     7  0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000
##     8  0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000
##     9  0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000
##     10 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.0000000 0.0000000
##       index
## states        85        86        87         88         89        90        91
##     1  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##     2  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##     3  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##     4  0.2896262 0.2245815 0.1833945 0.09169725 0.05760973 0.0000000 0.0000000
##     5  0.7103738 0.7754185 0.8166055 0.50000000 0.37173982 0.2210419 0.1105209
##     6  0.0000000 0.0000000 0.0000000 0.40830275 0.57065045 0.4851704 0.3531061
##     7  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.2937877 0.3894791
##     8  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.1468939
##     9  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##     10 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##       index
## states         92         93         94         95        96        97
##     1  0.00000000 0.00000000 0.00000000 0.07461461 0.2008796 0.3380928
```

```
##     2  0.00000000 0.00000000 0.00000000 0.00000000 0.0373073 0.1213572
##     3  0.00000000 0.00000000 0.00000000 0.00000000 0.0000000 0.0000000
##     4  0.00000000 0.00000000 0.00000000 0.00000000 0.0000000 0.0000000
##     5  0.05964091 0.02982045 0.00000000 0.00000000 0.0000000 0.0000000
##     6  0.25018916 0.15491503 0.09376581 0.00000000 0.0000000 0.0000000
##     7  0.40072459 0.32545687 0.24382138 0.00000000 0.0000000 0.0000000
##     8  0.28944535 0.34508497 0.34034555 0.00000000 0.0000000 0.0000000
##     9  0.00000000 0.14472267 0.24861066 0.59824088 0.2991204 0.1524031
##     10 0.00000000 0.00000000 0.07345659 0.32714452 0.4626927 0.3881469
##       index
## states        98        99       100
##     1  0.0000000 0.0000000 0.0000000
##     2  0.7910543 0.3955271 0.3063598
##     3  0.2089457 0.5000000 0.6936402
##     4  0.0000000 0.1044729 0.0000000
##     5  0.0000000 0.0000000 0.0000000
##     6  0.0000000 0.0000000 0.0000000
##     7  0.0000000 0.0000000 0.0000000
##     8  0.0000000 0.0000000 0.0000000
##     9  0.0000000 0.0000000 0.0000000
##     10 0.0000000 0.0000000 0.0000000
```

```
print(smoothed)
```

```
##       index
## states         1          2         3         4         5         6         7
##     1  0.1962687 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     2  0.4641791 0.39253731 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     3  0.3395522 0.53582090 0.7850746 0.4985075 0.2119403 0.0000000 0.0000000
##     4  0.0000000 0.07164179 0.2149254 0.5014925 0.7880597 0.7947761 0.5149254
##     5  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.2052239 0.4850746
##     6  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     7  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     8  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     9  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     10 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##       index
## states         8         9        10        11        12        13         14
##     1  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     2  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     3  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     4  0.2350746 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     5  0.7649254 0.8776119 0.7104478 0.5432836 0.3761194 0.2089552 0.04179104
##     6  0.0000000 0.1223881 0.2895522 0.4567164 0.6238806 0.7910448 0.95820896
##     7  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     8  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     9  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##     10 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000
##       index
## states        15        16        17 18 19        20        21         22
##     1  0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000 0.56783587
##     2  0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000 0.00000000
##     3  0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000 0.00000000
##     4  0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000 0.00000000
```

```
##     5  0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000 0.00000000
##     6  0.2813433 0.0000000 0.0000000  0  0 0.0000000 0.0000000 0.00000000
##     7  0.7186567 0.5208955 0.0000000  0  0 0.0000000 0.0000000 0.00000000
##     8  0.0000000 0.4791045 0.7604478  0  0 0.0000000 0.0000000 0.00000000
##     9  0.0000000 0.0000000 0.2395522  1  1 0.5790867 0.1581734 0.02117803
##     10 0.0000000 0.0000000 0.0000000  0  0 0.4209133 0.8418266 0.41098610
##       index
## states         23        24        25        26        27        28        29
##     1  0.57908670 0.1906023 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     2  0.33620119 0.6624752 0.3388484 0.0000000 0.0000000 0.0000000 0.0000000
##     3  0.00000000 0.1469226 0.6611516 0.5294507 0.0000000 0.0000000 0.0000000
##     4  0.00000000 0.0000000 0.0000000 0.4705493 0.7200529 0.0000000 0.0000000
##     5  0.00000000 0.0000000 0.0000000 0.0000000 0.2799471 0.9106552 0.4837856
##     6  0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0893448 0.5162144
##     7  0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     8  0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     9  0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##     10 0.08471211 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##       index
## states         30        31       32        33        34        35        36
##     1  0.00000000 0.0000000 0.000000 0.0000000 0.0000000 0.3143614 0.8113832
##     2  0.00000000 0.0000000 0.000000 0.0000000 0.0000000 0.0000000 0.1886168
##     3  0.00000000 0.0000000 0.000000 0.0000000 0.0000000 0.0000000 0.0000000
##     4  0.00000000 0.0000000 0.000000 0.0000000 0.0000000 0.0000000 0.0000000
##     5  0.05691595 0.0000000 0.000000 0.0000000 0.0000000 0.0000000 0.0000000
##     6  0.94308405 0.1826605 0.000000 0.0000000 0.0000000 0.0000000 0.0000000
##     7  0.00000000 0.8173395 0.308405 0.0000000 0.0000000 0.0000000 0.0000000
##     8  0.00000000 0.0000000 0.691595 0.4341496 0.0000000 0.0000000 0.0000000
##     9  0.00000000 0.0000000 0.000000 0.5658504 0.5598941 0.0000000 0.0000000
##     10 0.00000000 0.0000000 0.000000 0.0000000 0.4401059 0.6856386 0.0000000
##       index
## states         37        38 39 40        41          42          43          44
##     1  0.00000000 0.0000000  0  0 0.0000000 0.00000000 0.00000000 0.00000000
##     2  0.93712773 0.4685639  0  0 0.0000000 0.00000000 0.00000000 0.00000000
##     3  0.06287227 0.5314361  1  0 0.0000000 0.00000000 0.00000000 0.00000000
##     4  0.00000000 0.0000000  0  1 0.3367347 0.08163265 0.01020408 0.00000000
##     5  0.00000000 0.0000000  0  0 0.6632653 0.51020408 0.21428571 0.04081633
##     6  0.00000000 0.0000000  0  0 0.0000000 0.40816327 0.55102041 0.36734694
##     7  0.00000000 0.0000000  0  0 0.0000000 0.00000000 0.22448980 0.48979592
##     8  0.00000000 0.0000000  0  0 0.0000000 0.00000000 0.00000000 0.10204082
##     9  0.00000000 0.0000000  0  0 0.0000000 0.00000000 0.00000000 0.00000000
##     10 0.00000000 0.0000000  0  0 0.0000000 0.00000000 0.00000000 0.00000000
##       index
## states         45        46        47        48        49 50 51 52 53         54
##     1  0.00000000 0.0000000 0.0000000 0.0000000 0.2142857  1  1  1  0 0.0000000
##     2  0.00000000 0.0000000 0.0000000 0.0000000 0.0000000  0  0  0  1 0.6065574
##     3  0.00000000 0.0000000 0.0000000 0.0000000 0.0000000  0  0  0  0 0.3934426
##     4  0.00000000 0.0000000 0.0000000 0.0000000 0.0000000  0  0  0  0 0.0000000
##     5  0.00000000 0.0000000 0.0000000 0.0000000 0.0000000  0  0  0  0 0.0000000
##     6  0.10204082 0.0000000 0.0000000 0.0000000 0.0000000  0  0  0  0 0.0000000
##     7  0.51020408 0.2040816 0.0000000 0.0000000 0.0000000  0  0  0  0 0.0000000
##     8  0.35714286 0.6122449 0.3571429 0.0000000 0.0000000  0  0  0  0 0.0000000
##     9  0.03061224 0.1836735 0.6428571 0.5714286 0.0000000  0  0  0  0 0.0000000
##     10 0.00000000 0.0000000 0.0000000 0.4285714 0.7857143  0  0  0  0 0.0000000
```

```
##       index
## states        55         56        57         58         59          60
##     1  0.0000000 0.00000000 0.0000000 0.00000000 0.00000000 0.000000000
##     2  0.2131148 0.04918033 0.0000000 0.00000000 0.00000000 0.000000000
##     3  0.7868852 0.49180328 0.1967213 0.00000000 0.00000000 0.000000000
##     4  0.0000000 0.45901639 0.5737705 0.44262295 0.14754098 0.029508197
##     5  0.0000000 0.00000000 0.2295082 0.45901639 0.52459016 0.327868852
##     6  0.0000000 0.00000000 0.0000000 0.09836066 0.29508197 0.491803279
##     7  0.0000000 0.00000000 0.0000000 0.00000000 0.03278689 0.144262295
##     8  0.0000000 0.00000000 0.0000000 0.00000000 0.00000000 0.006557377
##     9  0.0000000 0.00000000 0.0000000 0.00000000 0.00000000 0.000000000
##     10 0.0000000 0.00000000 0.0000000 0.00000000 0.00000000 0.000000000
##       index
## states         61        62       63 64 65 66        67        68        69
##     1  0.00000000 0.0000000 0.000000  0  0  0 0.0000000 0.0000000 0.0000000
##     2  0.00000000 0.0000000 0.000000  0  0  0 0.0000000 0.0000000 0.0000000
##     3  0.00000000 0.0000000 0.000000  0  0  0 0.0000000 0.0000000 0.0000000
##     4  0.00000000 0.0000000 0.000000  0  0  0 0.0000000 0.0000000 0.0000000
##     5  0.11147541 0.0000000 0.000000  0  0  0 0.0000000 0.0000000 0.0000000
##     6  0.49180328 0.2754098 0.000000  0  0  0 0.0000000 0.0000000 0.0000000
##     7  0.35409836 0.5639344 0.557377  0  0  0 0.0000000 0.0000000 0.0000000
##     8  0.04262295 0.1606557 0.442623  1  1  1 0.7293576 0.4587153 0.2521097
##     9  0.00000000 0.0000000 0.000000  0  0  0 0.2706424 0.5412847 0.6198167
##     10 0.00000000 0.0000000 0.000000  0  0  0 0.0000000 0.0000000 0.1280736
##       index
## states        70         71        72        73        74         75         76
##     1  0.0000000 0.00000000 0.1486251 0.3798196 0.7100975 0.62999238 0.37310554
##     2  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.28010267 0.58349094
##     3  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.00000000 0.04340351
##     4  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     5  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     6  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     7  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     8  0.1095410 0.03100902 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     9  0.5702750 0.39265968 0.1860541 0.0000000 0.0000000 0.00000000 0.00000000
##     10 0.3201841 0.57633129 0.6653208 0.6201804 0.2899025 0.08990495 0.00000000
##       index
## states        77        78        79       80         81        82         83
##     1  0.0000000 0.0000000 0.0000000 0.000000 0.00000000 0.0000000 0.00000000
##     2  0.8496018 0.6588282 0.4680546 0.277281 0.08650744 0.0000000 0.00000000
##     3  0.1503982 0.3411718 0.5319454 0.722719 0.91349256 0.5007362 0.00000000
##     4  0.0000000 0.0000000 0.0000000 0.000000 0.00000000 0.4992638 0.91496492
##     5  0.0000000 0.0000000 0.0000000 0.000000 0.00000000 0.0000000 0.08503508
##     6  0.0000000 0.0000000 0.0000000 0.000000 0.00000000 0.0000000 0.00000000
##     7  0.0000000 0.0000000 0.0000000 0.000000 0.00000000 0.0000000 0.00000000
##     8  0.0000000 0.0000000 0.0000000 0.000000 0.00000000 0.0000000 0.00000000
##     9  0.0000000 0.0000000 0.0000000 0.000000 0.00000000 0.0000000 0.00000000
##     10 0.0000000 0.0000000 0.0000000 0.000000 0.00000000 0.0000000 0.00000000
##       index
## states       84        85        86        87         88          89         90
##     1  0.000000 0.0000000 0.0000000 0.0000000 0.00000000 0.000000000 0.00000000
##     2  0.000000 0.0000000 0.0000000 0.0000000 0.00000000 0.000000000 0.00000000
##     3  0.000000 0.0000000 0.0000000 0.0000000 0.00000000 0.000000000 0.00000000
##     4  0.727136 0.5393072 0.3514783 0.1636494 0.04941021 0.008760675 0.00000000
```

```
##      5  0.272864 0.4606928 0.6485217 0.8363506 0.62291512 0.262300312 0.06529091
##      6  0.000000 0.0000000 0.0000000 0.0000000 0.32767467 0.728939013 0.52164364
##      7  0.000000 0.0000000 0.0000000 0.0000000 0.00000000 0.000000000 0.41306544
##      8  0.000000 0.0000000 0.0000000 0.0000000 0.00000000 0.000000000 0.00000000
##      9  0.000000 0.0000000 0.0000000 0.0000000 0.00000000 0.000000000 0.00000000
##      10 0.000000 0.0000000 0.0000000 0.0000000 0.00000000 0.000000000 0.00000000
##        index
## states         91         92         93        94        95        96        97
##      1  0.00000000 0.00000000 0.00000000 0.0000000 0.1394209 0.4777218 0.6763176
##      2  0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0506985 0.3236824
##      3  0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##      4  0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##      5  0.00783491 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##      6  0.18356766 0.03286686 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##      7  0.63504053 0.33340193 0.08550927 0.0000000 0.0000000 0.0000000 0.0000000
##      8  0.17355691 0.63373120 0.48355351 0.1761756 0.0000000 0.0000000 0.0000000
##      9  0.00000000 0.00000000 0.43093722 0.5576573 0.3048657 0.0000000 0.0000000
##      10 0.00000000 0.00000000 0.00000000 0.2661671 0.5557134 0.4715797 0.0000000
##        index
## states         98        99       100
##      1  0.00000000 0.0000000 0.0000000
##      2  0.91907941 0.6127196 0.3063598
##      3  0.08092059 0.3872804 0.6936402
##      4  0.00000000 0.0000000 0.0000000
##      5  0.00000000 0.0000000 0.0000000
##      6  0.00000000 0.0000000 0.0000000
##      7  0.00000000 0.0000000 0.0000000
##      8  0.00000000 0.0000000 0.0000000
##      9  0.00000000 0.0000000 0.0000000
##      10 0.00000000 0.0000000 0.0000000
```

```r
print(probablePath)
```

```
##   [1]  1  2  3  3  3  4  4  4  5  5  5  5  5  5  6  7  8  9  9  9 10  1  1  1  2
##  [26]  3  4  5  5  6  7  8  9 10  1  1  2  2  3  4  4  4  5  6  7  8  9 10  1  1
##  [51]  1  1  2  2  2  2  3  4  4  4  5  6  7  8  8  8  8  8  8  9 10  1  1  1  1
##  [76]  1  2  2  2  2  2  3  4  4  4  4  4  4  5  6  7  8  9 10  1  1  1  2  2  2
```

Question 4: Accuracy

```r
# accuracy filtered
filteredStates <- apply(filtered, 2, which.max)
filteredAcc <- mean(filteredStates == simulation$states)
print(filteredAcc)
```

```
## [1] 0.56
```

```r
#accuracy smoothed
smoothedStates <- apply(smoothed, 2, which.max)
smoothedAcc <- mean(smoothedStates == simulation$states)
print(smoothedAcc)
```

```
## [1] 0.69
```

```
#accuracy probable path
probablePathAcc <- mean(probablePath == simulation$states)
print(probablePathAcc)
```

```
## [1] 0.6
```

Question 5: Repeat with a different sample

```
set.seed(222222)
simulation <- simHMM(hmm, 100)
observations <- simulation$observation

filtered <- rep(0, 100)
smoothed <- rep(0, 100)

logAlpha <- forward(hmm, observations)
logBeta <- backward(hmm, observations)

# filtered
filtered <- apply(logAlpha, 2, function(col){
  col <- exp(col - max(col))
  col/sum(col)
})

#smoothed
smoothed <- apply(logAlpha + logBeta, 2, function(col){
  col <- exp(col - max(col))
  col/sum(col)
})

#probable path
probablePath <- viterbi(hmm, observations)

#prints
print(filtered)
```

```
##       index
## states   1         2         3         4         5 6   7         8   9  10
##      1  0.0 0.0000000 0.0000000 0.00000000 0.0000000 1 0.5 0.0000000 0.0 0.0
##      2  0.0 0.0000000 0.0000000 0.00000000 0.0000000 0 0.5 0.6666667 0.4 0.2
##      3  0.0 0.0000000 0.0000000 0.00000000 0.0000000 0 0.0 0.3333333 0.6 0.5
##      4  0.2 0.1111111 0.0000000 0.00000000 0.0000000 0 0.0 0.0000000 0.0 0.3
##      5  0.2 0.2222222 0.0000000 0.00000000 0.0000000 0 0.0 0.0000000 0.0 0.0
##      6  0.2 0.2222222 0.2857143 0.00000000 0.0000000 0 0.0 0.0000000 0.0 0.0
##      7  0.2 0.2222222 0.2857143 0.33333333 0.1739130 0 0.0 0.0000000 0.0 0.0
##      8  0.2 0.2222222 0.2857143 0.33333333 0.3478261 0 0.0 0.0000000 0.0 0.0
##      9  0.0 0.0000000 0.1428571 0.25000000 0.3043478 0 0.0 0.0000000 0.0 0.0
##      10 0.0 0.0000000 0.0000000 0.08333333 0.1739130 0 0.0 0.0000000 0.0 0.0
##       index
```

```
## states   11        12        13         14         15        16        17
##      1  0.00 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##      2  0.10 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##      3  0.35 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##      4  0.40 0.5172414 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##      5  0.15 0.3793103 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##      6  0.00 0.1034483 0.8235294 0.41176471 0.20588235 0.1272727 0.0000000
##      7  0.00 0.0000000 0.1764706 0.50000000 0.45588235 0.4090909 0.2864078
##      8  0.00 0.0000000 0.0000000 0.08823529 0.29411765 0.4636364 0.4660194
##      9  0.00 0.0000000 0.0000000 0.00000000 0.04411765 0.0000000 0.2475728
##      10 0.00 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000
##        index
## states        18         19         20         21         22         23
##      1  0.0000000 0.06189320 0.15591735 0.23891547 0.29314067 0.32325244
##      2  0.0000000 0.00000000 0.00000000 0.07942584 0.16575320 0.25018109
##      3  0.0000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
##      4  0.0000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
##      5  0.0000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
##      6  0.0000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
##      7  0.1432039 0.07160194 0.03694427 0.00000000 0.00000000 0.00000000
##      8  0.3762136 0.25970874 0.17094552 0.10590112 0.05514034 0.03006157
##      9  0.3567961 0.36650485 0.32310582 0.25167464 0.18618170 0.13156465
##      10 0.1237864 0.24029126 0.31308704 0.32408293 0.29978409 0.26494024
##        index
## states         24         25        26        27        28        29        30
##      1  0.29858430 0.57771993 0.4751136 0.3483813 0.8055814 0.8372274 0.5000000
##      2  0.29109211 0.00000000 0.3032373 0.3891755 0.0000000 0.0000000 0.4186137
##      3  0.12699945 0.00000000 0.0000000 0.1516187 0.0000000 0.0000000 0.0000000
##      4  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##      5  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##      6  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##      7  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##      8  0.00000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##      9  0.08204633 0.09482575 0.0000000 0.0000000 0.0000000 0.0000000 0.0000000
##      10 0.20127781 0.32745431 0.2216491 0.1108245 0.1944186 0.1627726 0.0813863
##        index
## states        31        32         33         34        35        36        37
##      1  0.3030242 0.1515121 0.07575604 0.03894004 0.0000000 0.0000000 0.0000000
##      2  0.4787903 0.3909072 0.27120966 0.17834694 0.0000000 0.0000000 0.0000000
##      3  0.2181855 0.3484879 0.36969758 0.32943846 0.0000000 0.0000000 0.0000000
##      4  0.0000000 0.1090928 0.22879034 0.30763412 0.5154361 0.2738525 0.1369263
##      5  0.0000000 0.0000000 0.05454638 0.14564044 0.3667307 0.4686974 0.3712750
##      6  0.0000000 0.0000000 0.00000000 0.00000000 0.1178333 0.2574501 0.3630737
##      7  0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.1287250
##      8  0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##      9  0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##      10 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##        index
## states        38       39        40         41 42  43   44    45     46
##      1  0.0000000 0.000000 0.0000000 0.0000000  0 0.0 0.00 0.125 0.2500
##      2  0.0000000 0.000000 0.0000000 0.0000000  0 0.0 0.00 0.000 0.0625
##      3  0.0000000 0.000000 0.0000000 0.0000000  0 0.0 0.00 0.000 0.0000
##      4  0.0992596 0.000000 0.0000000 0.0000000  0 0.0 0.00 0.000 0.0000
##      5  0.3684016 0.000000 0.0000000 0.0000000  0 0.0 0.00 0.000 0.0000
```

```
##     6  0.5323388 0.628535 0.0000000 0.0000000  0 0.0 0.00 0.000 0.0000
##     7  0.0000000 0.371465 0.7291473 0.0000000  0 0.0 0.00 0.000 0.0000
##     8  0.0000000 0.000000 0.2708527 0.7868732  1 0.5 0.25 0.125 0.0625
##     9  0.0000000 0.000000 0.0000000 0.2131268  0 0.5 0.50 0.375 0.2500
##     10 0.0000000 0.000000 0.0000000 0.0000000  0 0.0 0.25 0.375 0.3750
##       index
## states         47        48         49         50         51         52
##     1  0.38461538 0.0000000 0.00000000 0.21428571 0.33928571 0.36036036
##     2  0.00000000 0.0000000 0.00000000 0.00000000 0.10714286 0.22522523
##     3  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.05405405
##     4  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
##     5  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
##     6  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
##     7  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
##     8  0.03846154 0.1428571 0.07142857 0.03571429 0.01785714 0.00000000
##     9  0.19230769 0.8571429 0.50000000 0.28571429 0.16071429 0.09009009
##     10 0.38461538 0.0000000 0.42857143 0.46428571 0.37500000 0.27027027
##       index
## states         53 54  55   56        57         58        59         60
##     1  0.37837838  0 0.0 0.00 0.0000000 0.00000000 0.0000000 0.00000000
##     2  0.35135135  0 0.0 0.00 0.0000000 0.00000000 0.0000000 0.00000000
##     3  0.00000000  1 0.5 0.25 0.0000000 0.00000000 0.0000000 0.00000000
##     4  0.00000000  0 0.5 0.50 0.4285714 0.21428571 0.1363636 0.06818182
##     5  0.00000000  0 0.0 0.25 0.4285714 0.42857143 0.4090909 0.27272727
##     6  0.00000000  0 0.0 0.00 0.1428571 0.28571429 0.4545455 0.43181818
##     7  0.00000000  0 0.0 0.00 0.0000000 0.07142857 0.0000000 0.22727273
##     8  0.00000000  0 0.0 0.00 0.0000000 0.00000000 0.0000000 0.00000000
##     9  0.05405405  0 0.0 0.00 0.0000000 0.00000000 0.0000000 0.00000000
##     10 0.21621622  0 0.0 0.00 0.0000000 0.00000000 0.0000000 0.00000000
##       index
## states         61         62         63        64         65         66
##     1  0.00000000 0.00000000 0.00000000 0.0000000 0.06344086 0.14301075
##     2  0.00000000 0.00000000 0.00000000 0.0000000 0.00000000 0.03172043
##     3  0.00000000 0.00000000 0.00000000 0.0000000 0.00000000 0.00000000
##     4  0.03409091 0.00000000 0.00000000 0.0000000 0.00000000 0.00000000
##     5  0.17045455 0.10404624 0.00000000 0.0000000 0.00000000 0.00000000
##     6  0.35227273 0.26589595 0.00000000 0.0000000 0.00000000 0.00000000
##     7  0.32954545 0.34682081 0.40151515 0.2046332 0.00000000 0.00000000
##     8  0.11363636 0.22543353 0.37500000 0.3957529 0.33440860 0.16720430
##     9  0.00000000 0.05780347 0.18560606 0.2857143 0.37956989 0.35698925
##     10 0.00000000 0.00000000 0.03787879 0.1138996 0.22258065 0.30107527
##       index
## states         67        68         69         70         71         72
##     1  0.24760192 0.4039409 0.41862069 0.31763547 0.22517643 0.1224406
##     2  0.00000000 0.1627586 0.28334975 0.35098522 0.35345434 0.3146328
##     3  0.00000000 0.0000000 0.08137931 0.18236453 0.28194578 0.3455014
##     4  0.00000000 0.0000000 0.00000000 0.04068966 0.11791360 0.2174252
##     5  0.00000000 0.0000000 0.00000000 0.00000000 0.02150986 0.0000000
##     6  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.0000000
##     7  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.0000000
##     8  0.09322542 0.0000000 0.00000000 0.00000000 0.00000000 0.0000000
##     9  0.29226619 0.0000000 0.00000000 0.00000000 0.00000000 0.0000000
##     10 0.36690647 0.4333005 0.21665025 0.10832512 0.00000000 0.0000000
##       index
```

```
## states        73         74        75        76        77        78       79
##     1  0.0612203 0.04078331 0.0000000 0.0000000 0.0000000 0.0000000 0.000000
##     2  0.2185367 0.18636657 0.1159391 0.0000000 0.0000000 0.0000000 0.000000
##     3  0.3300671 0.36546507 0.2816593 0.2110327 0.2702315 0.2127419 0.106371
##     4  0.2814633 0.40738505 0.3944689 0.3588675 0.7297685 0.7872581 0.500000
##     5  0.1087126 0.00000000 0.2079326 0.3197357 0.0000000 0.0000000 0.393629
##     6  0.0000000 0.00000000 0.0000000 0.1103641 0.0000000 0.0000000 0.000000
##     7  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.000000
##     8  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.000000
##     9  0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.000000
##     10 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000 0.000000
##       index
## states        80        81         82       83        84        85        86
##     1  0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000 0.0000000
##     2  0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000 0.0000000
##     3  0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000 0.0000000
##     4  0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000 0.0000000
##     5  0.6942113 0.3471056 0.00000000 0.000000 0.0000000 0.0000000 0.0000000
##     6  0.3057887 0.5000000 0.51249834 0.360915 0.1804575 0.0000000 0.0000000
##     7  0.0000000 0.1528944 0.39500066 0.639085 0.5000000 0.0000000 0.0000000
##     8  0.0000000 0.0000000 0.09250099 0.000000 0.3195425 0.7194744 0.3597372
##     9  0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.2805256 0.5000000
##     10 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000 0.1402628
##       index
## states        87        88         89        90        91        92        93
##     1  0.0000000 0.0000000 0.17625159 0.0000000 0.0000000 0.2287772 0.3643886
##     2  0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.1143886
##     3  0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##     4  0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##     5  0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##     6  0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##     7  0.0000000 0.0000000 0.00000000 0.0000000 0.0000000 0.0000000 0.0000000
##     8  0.2949936 0.1474968 0.07374841 0.1564969 0.0000000 0.0000000 0.0000000
##     9  0.7050064 0.5000000 0.32374841 0.8435031 0.5424455 0.2712228 0.1356114
##     10 0.0000000 0.3525032 0.42625159 0.0000000 0.4575545 0.5000000 0.3856114
##       index
## states         94         95         96         97         98         99
##     1  0.37500000 0.32716158 0.25250852 0.18534486 0.00000000 0.00000000
##     2  0.23938862 0.31623781 0.32741321 0.30167868 0.28536342 0.14268171
##     3  0.05719431 0.15265702 0.23861129 0.29444927 0.34929135 0.31732739
##     4  0.00000000 0.00000000 0.07768413 0.16453874 0.26893646 0.30911391
##     5  0.00000000 0.00000000 0.00000000 0.00000000 0.09640876 0.18267261
##     6  0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.04820438
##     7  0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
##     8  0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
##     9  0.06780569 0.03490091 0.00000000 0.00000000 0.00000000 0.00000000
##     10 0.26061138 0.16904267 0.10378285 0.05398845 0.00000000 0.00000000
##       index
## states        100
##     1  0.00000000
##     2  0.00000000
##     3  0.00000000
##     4  0.44831974
##     5  0.35195254
```

```
##     6  0.16522971
##     7  0.03449801
##     8  0.00000000
##     9  0.00000000
##     10 0.00000000
```

```
print(smoothed)
```

```
##       index
## states     1    2   3    4 5 6         7         8         9       10         11
##     1  0.000 0.00 0.0 0.00 0 1 0.3549236 0.0000000 0.0000000 0.000000 0.0000000
##     2  0.000 0.00 0.0 0.00 0 0 0.6450764 0.7098472 0.1295416 0.000000 0.0000000
##     3  0.000 0.00 0.0 0.00 0 0 0.0000000 0.2901528 0.8704584 0.323854 0.0000000
##     4  0.000 0.00 0.0 0.00 0 0 0.0000000 0.0000000 0.0000000 0.676146 0.5181664
##     5  0.000 0.00 0.0 0.00 0 0 0.0000000 0.0000000 0.0000000 0.000000 0.4818336
##     6  0.125 0.00 0.0 0.00 0 0 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##     7  0.500 0.25 0.0 0.00 0 0 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##     8  0.375 0.75 0.5 0.00 0 0 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##     9  0.000 0.00 0.5 0.75 0 0 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##     10 0.000 0.00 0.0 0.25 1 0 0.0000000 0.0000000 0.0000000 0.000000 0.0000000
##       index
## states        12         13         14        15        16         17
##     1  0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.00000000
##     2  0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.00000000
##     3  0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.00000000
##     4  0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.00000000
##     5  0.7124788 0.00000000 0.00000000 0.0000000 0.0000000 0.00000000
##     6  0.2875212 0.90679117 0.55722744 0.2930745 0.1143324 0.00000000
##     7  0.0000000 0.09320883 0.42447025 0.5849100 0.5745282 0.48182938
##     8  0.0000000 0.00000000 0.01830232 0.1220154 0.3111394 0.44166667
##     9  0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.07650395
##     10 0.0000000 0.00000000 0.00000000 0.0000000 0.0000000 0.00000000
##       index
## states         18          19          20         21         22         23
##     1  0.00000000 0.001122856 0.005482182 0.01649058 0.03885964 0.07859995
##     2  0.00000000 0.000000000 0.000000000 0.00000000 0.00000000 0.00000000
##     3  0.00000000 0.000000000 0.000000000 0.00000000 0.00000000 0.00000000
##     4  0.00000000 0.000000000 0.000000000 0.00000000 0.00000000 0.00000000
##     5  0.00000000 0.000000000 0.000000000 0.00000000 0.00000000 0.00000000
##     6  0.00000000 0.000000000 0.000000000 0.00000000 0.00000000 0.00000000
##     7  0.28386317 0.145640928 0.054730814 0.00000000 0.00000000 0.00000000
##     8  0.52008070 0.501348488 0.420651883 0.30797678 0.15972776 0.05950536
##     9  0.18306426 0.305808047 0.413780686 0.48382149 0.50056371 0.43862395
##     10 0.01299187 0.046079682 0.105354435 0.19171115 0.30084889 0.42327074
##       index
## states        24        25        26        27        28         29         30
##     1  0.1430211 0.2394326 0.3751441 0.5501556 0.7251671 0.90017863 0.45927874
##     2  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.51565788
##     3  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     4  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     5  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     6  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     7  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     8  0.0000000 0.0000000 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
```

```
##     9  0.3199309 0.1403155 0.0000000 0.0000000 0.0000000 0.00000000 0.00000000
##     10 0.5370480 0.6202519 0.6248559 0.4498444 0.2748329 0.09982137 0.02506337
##       index
## states        31         32         33        34         35        36
##     1  0.1790412 0.03704301 0.00000000 0.0000000 0.00000000 0.0000000
##     2  0.5609072 0.36636108 0.13261547 0.0000000 0.00000000 0.0000000
##     3  0.2600516 0.48990807 0.51219260 0.3133893 0.00000000 0.0000000
##     4  0.0000000 0.10668784 0.32629731 0.5365199 0.60603655 0.2525152
##     5  0.0000000 0.00000000 0.02889462 0.1500908 0.35932712 0.6050503
##     6  0.0000000 0.00000000 0.00000000 0.0000000 0.03463633 0.1424345
##     7  0.0000000 0.00000000 0.00000000 0.0000000 0.00000000 0.0000000
##     8  0.0000000 0.00000000 0.00000000 0.0000000 0.00000000 0.0000000
##     9  0.0000000 0.00000000 0.00000000 0.0000000 0.00000000 0.0000000
##     10 0.0000000 0.00000000 0.00000000 0.0000000 0.00000000 0.0000000
##       index
## states         37        38        39        40 41 42        43        44
##     1  0.00000000 0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000
##     2  0.00000000 0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000
##     3  0.00000000 0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000
##     4  0.05050305 0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000
##     5  0.54775521 0.1874418 0.0000000 0.0000000  0  0 0.0000000 0.0000000
##     6  0.40174174 0.8125582 0.4582946 0.0000000  0  0 0.0000000 0.0000000
##     7  0.00000000 0.0000000 0.5417054 0.7291473  0  0 0.0000000 0.0000000
##     8  0.00000000 0.0000000 0.0000000 0.2708527  1  1 0.8461538 0.6923077
##     9  0.00000000 0.0000000 0.0000000 0.0000000  0  0 0.1538462 0.3076923
##     10 0.00000000 0.0000000 0.0000000 0.0000000  0  0 0.0000000 0.0000000
##       index
## states        45        46        47         48         49        50         51
##     1  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.2769231 0.58461538
##     2  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.09230769
##     3  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.00000000
##     4  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.00000000
##     5  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.00000000
##     6  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.00000000
##     7  0.0000000 0.0000000 0.0000000 0.00000000 0.00000000 0.0000000 0.00000000
##     8  0.5384615 0.3846154 0.2307692 0.07692308 0.01538462 0.0000000 0.00000000
##     9  0.4615385 0.6153846 0.7692308 0.92307692 0.43076923 0.1230769 0.00000000
##     10 0.0000000 0.0000000 0.0000000 0.00000000 0.55384615 0.6000000 0.32307692
##       index
## states        52 53 54        55        56         57        58         59
##     1  0.6153846  0  0 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000
##     2  0.3846154  1  0 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000
##     3  0.0000000  0  1 0.4733343 0.1589001 0.00000000 0.0000000 0.00000000
##     4  0.0000000  0  0 0.5266657 0.6288685 0.47670031 0.1801897 0.05377096
##     5  0.0000000  0  0 0.0000000 0.2122314 0.46660242 0.5930212 0.37925627
##     6  0.0000000  0  0 0.0000000 0.0000000 0.05669728 0.2267891 0.56697277
##     7  0.0000000  0  0 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000
##     8  0.0000000  0  0 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000
##     9  0.0000000  0  0 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000
##     10 0.0000000  0  0 0.0000000 0.0000000 0.00000000 0.0000000 0.00000000
##       index
## states          60         61         62         63         64         65
##     1  0.000000000 0.0000000 0.00000000 0.00000000 0.00000000 0.01487076
##     2  0.000000000 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
```

```
##     3  0.000000000 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
##     4  0.009926781 0.0000000 0.00000000 0.00000000 0.00000000 0.00000000
##     5  0.175376735 0.0496339 0.00000000 0.00000000 0.00000000 0.00000000
##     6  0.522972289 0.3504798 0.15221064 0.00000000 0.00000000 0.00000000
##     7  0.291724195 0.4703526 0.47981245 0.35074625 0.10157647 0.00000000
##     8  0.000000000 0.1295336 0.32066511 0.46410677 0.48188495 0.29802154
##     9  0.000000000 0.0000000 0.04731181 0.17317751 0.34591876 0.49151355
##     10 0.000000000 0.0000000 0.00000000 0.01196946 0.07061982 0.19559415
##       index
## states         66        67         68         69         70          71
##     1  0.06704444 0.2081906 0.37325219 0.53573120 0.53247794 0.432929943
##     2  0.00000000 0.0000000 0.05779732 0.16102773 0.30996416 0.441302438
##     3  0.00000000 0.0000000 0.00000000 0.01154944 0.04622476 0.118993982
##     4  0.00000000 0.0000000 0.00000000 0.00000000 0.00123565 0.006773637
##     5  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.000000000
##     6  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.000000000
##     7  0.00000000 0.0000000 0.00000000 0.00000000 0.00000000 0.000000000
##     8  0.08046548 0.0000000 0.00000000 0.00000000 0.00000000 0.000000000
##     9  0.46449268 0.2522632 0.00000000 0.00000000 0.00000000 0.000000000
##     10 0.38799739 0.5395462 0.56895049 0.29169162 0.11009748 0.000000000
##       index
## states         72         73         74        75        76        77
##     1  0.23388710 0.10400812 0.03035756 0.0000000 0.0000000 0.0000000
##     2  0.51147588 0.46362606 0.33655928 0.1690818 0.0000000 0.0000000
##     3  0.23166679 0.37289454 0.50387095 0.5857897 0.5798445 0.3327713
##     4  0.02297023 0.05947128 0.12921221 0.2451285 0.4201555 0.6672287
##     5  0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##     6  0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##     7  0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##     8  0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##     9  0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##     10 0.00000000 0.00000000 0.00000000 0.0000000 0.0000000 0.0000000
##       index
## states         78        79        80         81       82        83        84
##     1  0.00000000 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000
##     2  0.00000000 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000
##     3  0.08569814 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000
##     4  0.91430186 0.4028267 0.0000000 0.00000000 0.000000 0.0000000 0.0000000
##     5  0.00000000 0.5971733 0.7199553 0.26804809 0.000000 0.0000000 0.0000000
##     6  0.00000000 0.0000000 0.2800447 0.65096496 0.654167 0.2054618 0.0000000
##     7  0.00000000 0.0000000 0.0000000 0.08098694 0.345833 0.7945382 0.5692804
##     8  0.00000000 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.4307196
##     9  0.00000000 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000
##     10 0.00000000 0.0000000 0.0000000 0.00000000 0.000000 0.0000000 0.0000000
##       index
## states         85        86       87        88       89         90        91
##     1  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     2  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     3  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     4  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     5  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     6  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     7  0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.0000000
##     8  0.93309899 0.7615155 0.589932 0.4183485 0.246765 0.07518146 0.0000000
```

```
##     9  0.06690101 0.2384845 0.410068 0.5816515 0.753235 0.92481854 0.4804023
##     10 0.00000000 0.0000000 0.000000 0.0000000 0.000000 0.00000000 0.5195977
##       index
## states        92         93         94          95         96         97
##     1  0.2663347 0.47365646 0.44995370 0.290535304 0.13770418 0.03886032
##     2  0.0000000 0.11764476 0.33510958 0.456356013 0.39281672 0.25300579
##     3  0.0000000 0.00000000 0.03758097 0.194877301 0.39037621 0.43214984
##     4  0.0000000 0.00000000 0.00000000 0.000000000 0.06778341 0.27598405
##     5  0.0000000 0.00000000 0.00000000 0.000000000 0.00000000 0.00000000
##     6  0.0000000 0.00000000 0.00000000 0.000000000 0.00000000 0.00000000
##     7  0.0000000 0.00000000 0.00000000 0.000000000 0.00000000 0.00000000
##     8  0.0000000 0.00000000 0.00000000 0.000000000 0.00000000 0.00000000
##     9  0.1801509 0.05617608 0.01355974 0.001937106 0.00000000 0.00000000
##     10 0.5535145 0.35252270 0.16379601 0.056294276 0.01131949 0.00000000
##       index
## states        98         99        100
##     1  0.0000000 0.00000000 0.00000000
##     2  0.1021118 0.00000000 0.00000000
##     3  0.3749614 0.22709891 0.00000000
##     4  0.3849348 0.44244167 0.44831974
##     5  0.1379920 0.26146341 0.35195254
##     6  0.0000000 0.06899601 0.16522971
##     7  0.0000000 0.00000000 0.03449801
##     8  0.0000000 0.00000000 0.00000000
##     9  0.0000000 0.00000000 0.00000000
##     10 0.0000000 0.00000000 0.00000000
```

```
print(probablePath)
```

```
##   [1]  6  7  8  9 10  1  1  2  2  3  4  5  6  6  7  8  9 10  1  1  1  1  1  1  1
##  [26]  1  1  1  1  1  1  1  2  3  4  4  4  5  6  7  8  8  8  8  8  8  8  9 10  1
##  [51]  1  1  2  3  3  3  4  4  5  6  7  8  9 10  1  1  1  1  1  1  1  1  1  1  2
##  [76]  3  3  3  4  5  5  6  6  7  8  8  8  8  8  9 10  1  1  1  1  1  1  2  3  4
```

```
# accuracy filtered
filteredStates <- apply(filtered, 2, which.max)
filteredAcc <- mean(filteredStates == simulation$states)
print(filteredAcc)
```

```
## [1] 0.42
```

```
#accuracy smoothed
smoothedStates <- apply(smoothed, 2, which.max)
smoothedAcc <- mean(smoothedStates == simulation$states)
print(smoothedAcc)
```

```
## [1] 0.46
```

```
#accuracy probable path
probablePathAcc <- mean(probablePath == simulation$states)
print(probablePathAcc)
```

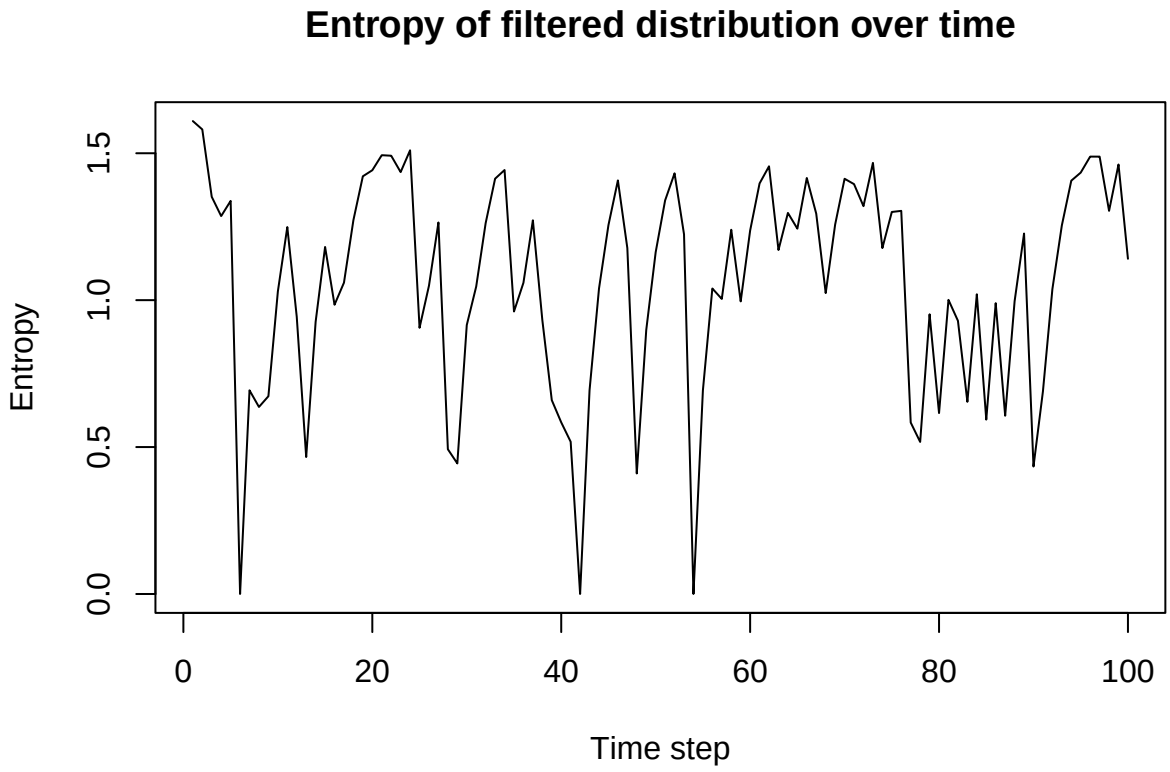
```
## [1] 0.39
```

ANSWER: The smoothing is in general better than filtered since smoothing has access to the entire state sequence as opposed to filtered. Most probable path will in general be outperformed by smoothing because smoothing is maximizing P(zt | x0:T) separately for every t. Viterbi does it as a whole, Viterbi needs to respect the internal consistency of the transition model and cannot jump around implausibly among the states.

Question 6: Entropy of the filtered distributions

```r
filtered_entropy <- apply(filtered, 2, entropy.empirical)

plot(filtered_entropy, type = "l",
     xlab = "Time step", ylab = "Entropy",
     main = "Entropy of filtered distribution over time")
```



ANSWER: No. If it were always true, entropy would decrease monotonically over time. Instead the plot fluctuates up and down: some new observations are informative and sharpen the filtered distribution (entropy drops), while others are ambiguous (e.g. they fall in the overlapping part of the device's reporting window) and increase uncertainty about the robot's current sector, even at late time steps. Since filtering is first-order Markov, only recent evidence matters for p(z_t | x_0:t), old accumulated certainty isn't "banked", so uncertainty can rise again regardless of how large t is.

Question 7: Probabilities for time step 101

```r
filtered_t100 <- filtered[, 100]         # p(z_100 | x_0:100), length-10 vector

probs_101 <- as.numeric(filtered_t100 %*% transProbs)
names(probs_101) <- states

print(probs_101)
```

```
##          1          2          3          4          5          6          7
## 0.00000000 0.00000000 0.00000000 0.22415987 0.40013614 0.25859113 0.09986386
##          8          9         10
## 0.01724900 0.00000000 0.00000000
```